Kant and Conceptual Semantics

A Sketch

Carl J. Posy

I believe that Kant has a view of predication and truth that straddles the present divide between reference based (broadly Fregean) semantic theories and those theories (discussed by Michael Dummett and others) which base truth on "warranted assertability" in one form or another.

And I believe that Kant inherited this view from the Leibnizian "conceptual-containment" theory of truth. Truth should be assigned, says Leibniz, "to the relationship amongst the objects of the ideas, by virtue of which one idea is or is not included within the other.[1] So, for him, semantics is based on ideas; it is a relation of conceptual inclusion.[2] But that conceptual arrangement is itself a relation among objects.

As I said, I believe that Kant also takes truth both in a Fregean referential way and in a conceptualist fashion. And I shall suggest in what follows that this dual theory helps explain the delicate interaction in the work of both Leibniz and Kant among semantics, ontology and logic. But I shall also claim that Kant radically changes Leibniz's account of concepts and their relations. Indeed, I shall claim that he replaces Leibniz's conceptual inclusion model of truth with a semantics that is closer to Dummett's account of assertability, and that noting this revision will explain some hitherto anomalous Kantian points.

There is, I know, an apparant anachronism in speaking of Leibniz and Kant as Fregean realists and as Dummettian anti-realists. I am, however, not bothered by this for two reasons.

For one thing, the link that Dummett has explored between semantic anti-realism and intuitionistic logic can be very helpful in explaining some controversial Kantian themes and arguments. There is no anachronism in saying that Kant exploited insights and connections which we have begun to make rigorous only recently.

But secondly, I do believe that the differences between Leibniz and Kant on the one hand and modern writers on the other are as important as the similarities. In particular, I shall argue that the differences between the two groups of writers reflect important differences in attitude about the relation of semantics, logic and ontology. These differences will also shed some light on the recently debated question about the extent to which Frege himself is an idealist in the mold of Kant and Leibniz.

I will not in this paper argue for these points in detail. I will rather, as my subtitle suggests, simply sketch how such an argument might go, by focusing on what I call "elementary predications": simple subsumptions of subject under predicate. Syntactically these judgments lie at the base of recursive clauses for compound statements. They are semantically primitive as well: The general theory of truth for these elementary predications is the prime influence on the truth conditions for compound statements.

I will set the stage in Part I by briefly explaining some of Leibniz's ideas about semantics and ontology and the effect of these ideas on his logic. Then, in Part II, I will indicate how Kant has transformed the central Leibnizian notion — the notion of a concept — and how this transformation effects Kant's views of semantics, ontology and logic. I will show there that, despite the power of using intuitionistic logic to interpret Kant, Kant is not an assertabilist. Finally, in the third part I will discuss (again briefly) the question of the parallels between the views of Kant and Leibniz on the one hand and modern thinkers on the other concerning two other predication-related issues.

I. Leibniz

The key to Leibniz's theory of elementary predication and to his general semantics is his notion of a concept. The key to his ontology is the notion of an individual concept. I will describe these briefly, and then explain their relation to Leibniz's view of logic.

Topoi **10**: 67–78, 1991.
© 1991 *Kluwer Academic Publishers. Printed in the Netherlands.*

i. *Concepts: general and individual*

A concept, according to Leibniz, is a "disposition to think of the object when the occasion arises."[3] It is, thus, a functional entity which takes mental occasions into thoughts. The thoughts, in particular, are elementary judgments which analyze the given concept into more general ones (*Merkmale*). Thus, for instance, *Man* is analyzed into *rational, animal,* etc. Concepts are resolvable ultimately into maximally general "primitive ingredients" (elementary predicates). There is, thus, a set of mutually independent basic concepts from which all the rest are constructible.[4] One can combine these basic predicates and their negations into increasingly specific compound concepts. One can also serially order these combinations, thus creating a sequence of "stages." But so long as some elementary predicate or other is not "determined" (i.e., affirmed or negated) by the combination the result is still a general concept.

At the extreme end of this spectrum, however, lie the individual concepts — representing individual objects (*individual substances*). For Leibniz an individual concept is *completely determined.* It is a set containing each primitive predicate or its denial for every stage of the object.[5] When you entertain such a concept you are projecting mentally; you are imagining yourself experiencing the object at those times. To use a modern simile, an individual concept is like a cable of mental optic fibers. There is one fiber for each elementary predicate, and each fiber contains information about the stages at which its simple predicate is thought to be true of the object in question. The fiber is what Leibniz called a "law" for our imaginings; and the concept is the collection of these simple laws.

According to Leibniz an object is unified over time simply by virtue of its complete individual concept. And the same goes for unity at a moment: A cross-section of the concept at any moment must contain information about the status of every possible elementary predicate.

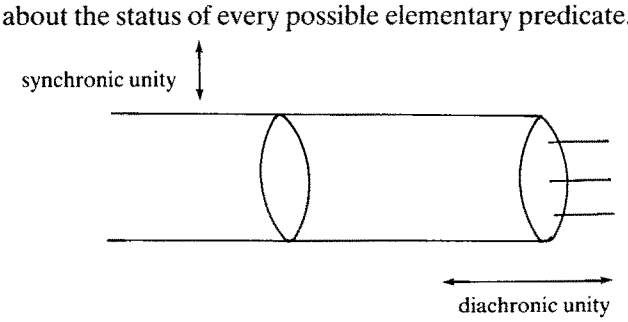

Diagram 1.

A concept which lacks this completeness in either dimension would not provide this unity, and thus would not be able to represent an object.

ii. *Conceptual semantics*

For Leibniz an elementary predication will be true when the analysis of its subject concept produces the predicate concept. In Leibniz's words, the subject concept *contains* the predicate. This famous containment theory of predication is the heart of the Leibnizian notion of truth, of his "conceptual semantics."[6]

Since analysis proceeds from the specific to the general, individual concepts can never occur as predicates in a predication. But when they occur as subjects, then they have no special semantic status. A singular predication (one whose subject is an individual concept) is, like all predications, true in virtue of the analysis of its subject concept.[7]

iii. *Perceptions and truth*

When you or I entertain the concept of some object we project and consider various unactualized mental states. Should that projection be complete, so that the whole content of the concept is arrayed before the mind, we would then have what Leibniz calls "intuitive" knowledge, knowledge of all the details of the represented object. For us humans, says Leibniz, this almost never occurs.[8] God may have intuitive knowledge. We, however, have to make do with sensory perception, which, on Leibniz's view, is merely confused thought. Individual perceptions are, to be sure, still concepts; but they are deeply inferior ones. For an individual perception by itself is not subject to further conceptual analysis. Moreover it is degenerate as a function — it has a single input, right now; and a single output, the present sensory state. Similarly, empirical concepts that we create from repeated individual perceptions inherit this inferiority. They cannot have more in them than experience has provided, so they are not fully analyzable. They always leave out details. Even when they purport to be individual concepts, they are still limited by the limits of experience, and thus they are still incomplete both synchronically and diachronically. They do not supply the unity of a true object and thus they fall short of reality.

But our epistemological weakness has no bearing on matters of truth. Only these complete ideas determine an object, and thus truth. And in this function it matters not whether they happen to be entertained (or even entertainable) by some person or other. The upshot is that those ideas which determine truth are epistemologically transcendent. And truth, for Leibniz, even though it is conceptual, is still inhuman.

It would be a mistake to use these grounds to impute a general ontological idealism to Leibniz. As far as he is concerned: Of course there are objects which are not concepts. Monads are prime examples. It is simply that the object has no properties other than those contained in its individual concept, the properties that ultimately figure into the truth of judgments about the object. God's judgments, if not our own. It is to this extent, and this extent only that truth of a judgment consists in the agreement of its predicate with an object.[9]

iv. *Logic*

The famous Leibnizian logical principles — the identity of indiscernibles, the principle of sufficient reason, the law of excluded middle — are ontological principles about the most general properties of objects.[10] They indeed follow to a large extent from Leibniz's conception of a fully determinate individual concept.[11] Thus, for instance, the principle of sufficient reason says that all changes in an object's status are already present in the object's individual concept, as are their causes.[12]

Since numerical difference between objects must have a sufficient reason, the identity of indiscernibles follows from the principle of sufficient reason and thus from completeness as well.[13] And the law of the Excluded Middle says that, for any given individual concept and any simple or complex predicate, either the predicate belongs to the concept or its negation does. This law is a clear corollary of the completeness of individual concepts. Indeed, since, according to Leibniz, a full individual concept will include the relation of the individual with all other elements and aspects of the universe, the law of excluded middle will hold across the board — for all judgments about any aspect of reality. Our own current approach to logic centers more on the logical properties of judgments than of objects, and on truth values rather than on causation or internal relations. But it is easy to see that these Leibnizian principles — governed as they are by the completeness of individual concepts — will dictate what we would now call a generally bivalent semantics.

II. Kant

Clearly Kant's debt to Leibniz runs very deep.[14] Like Leibniz, Kant views concepts as functions that link mental representations to other mental representations. (Though the operative word is now "rule".) And he follow's Leibniz's general containment model of subsumption. An object, Kant says, is subsumed under a concept when the representation of that object is "homogeneous" with the predicate concept.[15] Finally — though we shall see that this is delicate — Kant also holds that individual concepts (and their objects) are subject to a principle of complete determination.[16]

But Kant also revolts in a profound way against Leibniz's model of predication. He injects sensation (or more precisely human, spacio-temporal perception) into the notion of truth. This semantic humanism effects his view about each of our Leibnizian topics: concepts, objects, elementary truth and logic.

i. *Intelligible and empirical representations*

Kant transforms Leibniz's theory of concepts in two important ways. First, he distinguishes between "intelligible" and "empirical" states of mind. And then he divides the empirical mental states between concepts and intuitions. (That is why he introduces the generic notion of a "representation", which is neutral between concept and intuition.)

Intelligible representations — representations which do not arise directly from our sensory experience — are just Leibnizian concepts. They are, you may say, God's dispositions to perceive. In particular, a singular intelligible concept is a fully complete presentation of an object. Kant mentions the concept of the "perfect man"[17] and the concept of God Himself[18] as examples of singular intelligible concepts. To be sure, this presumed completeness is still (as Leibniz said) beyond our human grasp. And these complete concepts do not enter into the determination of truth. Nor can we assert — at least not in empirical discourse — that the objects of these intelligible concepts exist.

Empirical representations, by contrast, are linked to experience. Some of them actually arise *a posteriori*

from experience. But all of them, whether derived or *a priori*, are limited by the forms of human experience and by our finite natures. Thus they are in general, as Leibniz insisted, incomplete. Nonetheless a central theme of the *Critique of Pure Reason* is that these humanly limited representations do have semantic weight, and they can determine objects. Indeed, from the perspective of scientific discourse, only these empirical concepts can enter into true judgments.[19]

ii. *Perceptions and concepts*

Kant divides empirical representations, as did Leibniz, into general predicates and singular representations. But he then divides the notion of a singular representations quite sharply between the notion of a *perception* and the notion of an individual *concept*, transforming what was for Leibniz a difference in degree into a difference in kind. Kant agrees that perceptions are degenerate as functions and are not subject to further conceptual analysis. But, unlike Leibniz, he does not conclude that these perceptions are therefore defective concepts. Instead he concludes that perceiving is a radically different mental activity than thinking or conceiving.

In the *Critique of Pure Reason* Kant speaks of perceptions as "empirical intuitions".[20] And here, I suggest, he uses the word advisedly. *Perception* is for Kant what *intuition* is for Leibniz: A perception is a fully adequate conscious representation of a *single object*. Its degeneracy as a function is, according to Kant, a virtue. A virtue he calls "immediacy." Indeed, Kant argues, because a perception relates immediately to its object, having a perception suffices to show that the object exists.

By contrast, singular empirical *concepts* are for Kant, as they were for Leibniz, just special cases of general concepts. Each such concept is a collection of *Merkmale* which (if it applies to anything) applies to only one object. Such a concept is not related to its object "immediately." It picks out the object by means of general features, features which may be shared with other objects.[21]

To achieve "immediacy" one must have, as I said, an intuition, a perception. Though for empirical objects this must contain sensation, it is not a mere sense datum. It is a representation which frames the momentary sensation with a concept. The immediacy and guarantee of existence is provided by the sensation. You would have no perception of the page before you, if you had no sensations of it. And of course there would be no sensation if there were no external source for those sensations. That is a fact of life for us humans. It is the fact that connects empirical intuition with existence.

But — and on this Kant agrees with Leibniz — sensation alone does not establish the presence of an object. Like Leibniz, he says that raw sensation lacks unity. And, again like Leibniz, he believes that the object-constituting unity comes from the projection involved in applying a concept. But then Kant differs. He employs a different notion of concept, a different projection, and ultimately a different sort of unity in his theory of empirical objects.

iii. *The representation of an object*

Suppose, for instance, that you judge there to be a white page before you. Then in order to apply the relevant concepts to the "manifold of sensations" with which you are presented, you must group together the white sensations from the various parts of the page region of your manifold. And you must distinguish these not only from the black-letter sensations but also from the white sensations that might be part of the background. Moreover for the judgment to be "objective" (in order, that is, for it to apply to an object) you must also suppose that there can be further glimpses of this same object by yourself and others. And you must project that these further glimpses will preserve the same organization (black letters against white page etc.), while perhaps increasing your information. Without these assumptions, your grouping would be nothing but a subjective combination of momentary impressions. So for Kant, just like for Leibniz, applying a concept involves an element of projection to as yet unrealized mental states.

But this must now be a "fuzzy" projection. For recall we are dealing with Kantian empirical concepts. Neither the general concept "page," nor the individual concept "page 71 of Posy's paper," will tell you what colors you will find when you examine its other side. Of course you do believe that the underside will have some color. That belief is indeed carried by the concepts. But the perceptions which can supply that color are out of your control. They are "receptive." That is part of the same fact of life. So the rule must fan out all the possibilities before you.

In the words of my earlier image, Kant has frayed the

strands in the Leibnizian cable in both the synchronic and diachronic dimensions. Indeed instead of the unidirectional Leibnizian strand, Kant must model the representation of an individual object (even of its synchronic aspects) as a branching, tree-like structure. The base includes what we know; the branches what we might come to know both about the present state and about other temporal states.

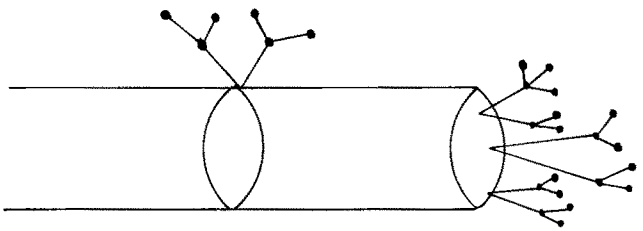

Diagram 2. The horizontal tree represents potential later evidence about future states of the object. The vertical tree represents possible new evidence about the object's present state.

For Kant, this branching projection provides all we can have of the unity needed to relate a judgment to its object.[22]

iv. *Semantics*

The projection to future glimpses makes the judgment objective. But objectivity is not truth. The judgment is not true unless the assumptions built into this projection (assumptions about those aspects of the organized manifold which will persevere through increased information) are justified.

Indeed, Kant's distinction between empirical and intelligible concepts induces a parallel distinction between two kinds of elementary predication and thus two kinds of truth: When a simple predication involves a (Leibnizian) intelligible concept then the subsumption is just the usual Leibnizian containment.

When, on the other hand, we are dealing with empirical concepts and the representations of empirical objects, then the subsumption has a more subtle meaning. The object, t, is subsumed under the predicate, P, if actual perceptions or the weight of indirect evidence justifies us in including P in the projections that are part of the individual representation of t. In that case the judgment Pt is true.

These two different models of subsumption (predication) generate two different types of truth: intelligible truth and empirical truth. The Leibnizian semantics, the semantics of intelligible truth, has an important place in Kant's ethical theory.[23] But, as I said, Kant's basic point in the *Critique of Pure Reason* is the claim that the only admissible notion of scientific truth is the empirical theory that I have just sketched.

The mistake of his predecessors, Kant claims, is to miss this distinction: to separate questions of knowledge and truth, even in empirical discourse, and thus to assume unquestioningly the full Leibnizian semantics. They are led naturally to the fallacies chronicled in the "Dialectic" of the *Critique of Pure Reason.*

v. *Logic and assertabilism: the ladder to intuitionism*

Singular intellectual concepts are in principle complete. So, were they applicable they would support Leibniz's ontological principles: sufficient reason, identity of indiscernibles and excluded middle. Similarly the semantics of "intellectual judgments" would be bivalent and essentially classical.

It is not, however, according to Kant, a semantics that holds valid for empirical truth. Leibniz's informal principles can no longer claim validity for empirical discourse. Kant insists, for example, that perception alone is a perfectly fine criterion for diversity.[24]

Indeed, empirical predications ought not to be bivalent. It seems, in fact, that for empirical judgments Kant is well on the way to the "intuitionistic" logic that Dummett connects with the well known assertabilism. Assertabilism, briefly put, holds a proposition to be true only when it is constructively proved, and false only when it is (again constructively) refuted. As a consequence the assertabilist theory of meaning yields a logic which invalidates bivalence, excluded middle and many other classical laws.[25]

It is tempting to make this association (of Kant's empirical logic with intuitionistic logic) because both equate empirical truth with empirical evidence. Moreover, the structure of Kant's branching empirical concepts looks much like the model structures that are used for the semantics of intuitionistic logic. And this is no accident. In the tree-like intuitionistic semantics the nodes represent "evidential situations". The branches that emanate from a given node represent — exactly as they do in our reconstruction of Kant's theory of

empirical concepts — the alternative possible extensions of the evidence at that node.

There are, as I mentioned at the start, also exegetical reasons to assume that Kant favored intuitionistic logic for empirical discourse. This assumption helps us, for instance, understand Kant's refutation of Transcendental Realism in the first antinomy, the one concerned with the age and spatial extent of the universe. There the transcendental realist is supposedly trapped by arguments which support the contradictory conclusions that the physical world is both finitely and infinitely extended into space; while the idealist escapes by holding both alternatives to be false. We can symbolize the finite extent of the world by "$\exists x \forall y(y \neq x \rightarrow Fxy)$" (where x and y range over occupied spatial regions, and Fxy says that x is further from some fixed geographic place than is y). And we can symbolize spatial infinity by "$\forall x \exists y Fyx$". Then to interpret Kant's *reductio* we need only note that when we add the axioms for spacial ordering[26] the disjunction "$[\exists x \forall y(y \neq x \rightarrow Fxy) \vee \forall x \exists y Fyx]$" is classically but not intuitionistically valid.

Indeed, a simple intuitionistic model structure which falsifies both disjuncts looks like the chain depicted in Diagram 3. Each node contains finitely many occupied spatial regions and each contains one which is further than the furthest one in its predecessor. This gives a good reading to the Kantian notion of "regulativity," the sense in which the apparent infinity of the world in space is just continuability "*ad indefinitum*" of the search for ever more remote regions.

This in turn appears to give a good reading of the principle of complete determination for empirical objects. For that too seems to be a regulative principle.[27] And that comes down to $\neg\neg(Pa \vee \neg Pa)$, which is a theorem of intuitionistic logic.

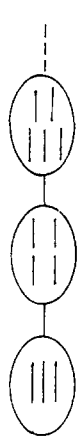

Diagram 3.

vi. *Removing the ladder: Kant is not an assertabilist*

Nevertheless, as I said at the outset, I want to resist these straightforward equations. Kant's empirical logic is not, I believe, standard Intuitionistic Logic; and Kant's theory of truth is not an early assertabilism. I have three reasons, stemming from considerations of what I call Peircianism, epistemology and ontology.

First Peircianism: Even on an epistemic theory of truth it is possible to distinguish the semantic factor from the epistemic factor. Knowledge is temporally relative, and truth need not be. I call this separation the Peircian move. Peirce argued strongly that in an epistemic semantics we must take truth as independent of knowledge at any particular time.[28] And for Kant as well as for Peirce, those judgments are true which eventually come to be and remain justifiably assertable.[29]

Indeed in the very same passage in which Kant admits that a concept may be incomplete, he goes on to say that an object (and thus an individual concept) is nevertheless "fully determinable" with respect to any given pair of contradictory predicates.[30] That is, for any predicate P and object t, $(Pt \vee \neg Pt)$ holds.[31] The key lies, I believe, in the fact that the existence of t is already established by perception. Excluded middle holds for judgments of the form Pt because (unlike the case of existential judgments) in this elementary case all the material for the necessary determination is already at hand. Thus, for elementary singular predication, we may say that the semantics and the logic are Leibnizian, even though the concepts themselves are not.[32] And the non-classical logic, which saved Kant's reasoning in the "Antinomy," will hold only at the level of quantification.

But what, you may ask, of the principle of complete determination and its regulativity? Is that not an application of intuitionistic logic to atomic predicates? The answer in brief is, no. When you look closely you will find that the regulativity which attaches to the principle of complete determination actually concerns the set of all possible predicates in general, and not the determination (i.e., bivalence) of attributing a particular known predicate to a particular known object. For any *given* t and p Kant does require $(Pt \vee \neg Pt)$ to be a logical law. He questions only $(\forall P)(\forall x)(Px \vee \neg Px)$. The problem that concerns Kant in these passages is not bivalence in an evidential theory of truth. The problem is whether there is a legitimate entity which is the set of all possible predicates.[33]

Secondly, regarding ontology, assertabilism is (or should be) ontologically neutral. In particular, since he

does not base truth on objects, the assertabilist has no right to deduce a phenomenalist ontology from his phenomenalist conception of truth. Dummett admits as much, and distinguishes assertabilism from ontological phenomenalism.[34] But Kant's anti-realism is explicitly ontological: Empirical objects are called "appearances". "[T]hey are nothing but representations, and cannot exist outside our mind." [A492/B520].

Kant's reasoning here is, once again, straigthforwardly Leibnizian: The individual representation of an empirical object contains spatio-temporal predicates which are, according to Kant, peculiarly human. Since, following the Leibnizian line, the individual representation dictates all there is to the object, it follows that the object is itself peculiarly human.[35]

It is tempting to separate Kant from the assertabilists on yet another ground. For Kant does speak of truth as the "agreement of a cognition with its object."[36] This sounds like Frege's referentialism, and it is scarcely what we would expect from an assertabilist. But let me inject a note of interpretative caution: Kant is simply again making a Leibnizian point: The properties of an object, including the property of "agreement" with any given cognition, are all carried in its representation. When we are talking about the truth of a simple subject/predicate judgment, then we may assume that the "cognition" in question is the predicate concept, and that the agreement in question is — as Leibniz says — the agreement of the subject and predicate concepts. For Kant the nature of these concepts (intelligible or empirical) determines whether that agreement is to be a simple Leibnizian inclusion or the more elaborate projections from incomplete human data that I have described above. That is why in the same passage he disparages the definition as merely "nominal." But in neither case is Kant embracing the non-conceptual (extensional) theory of direct reference which lies at the heart of Fregean realism.

I do not want to defend Kant's views here, nor Frege's, nor Dummett's. I simply want to point out that Kant and Dummett are fighting different enemies. Kant is fighting Leibniz. And to a large extent, as I have tried to show, this fight is in Leibniz's own semantic and ontological arena. Dummett is fighting Frege, again, I believe, on Frege's own grounds.

III. Two Current Issues

This raises the question I mentioned initially of just how much overlap there is between the modern paradigm of predication — a paradigm whose roots lie in Frege and Russell — on the one hand and the framework within which Leibniz and Kant worked on the other.

Two things seem clear. First, Frege and all the moderns differ from Leibniz on the question of whether singular predication — and with it singular terms and singular concepts — constitute separate linguistic categories. Nowadays expressions of the form *Pt* form the base of all syntactic and semantic recursions.[37] But for Leibniz the difference between general and singular terms (and, for that matter, general and singular concepts) is at most a matter of degree.[38] For him, judgments of the form *All A's are B's* are the most basic predications; and the conceptual-containment relations that underly these judgments are the most basic truth conditions. Kant — if we are looking for partisanship here — sides with the moderns regarding empirical discourse. Empirical truth, for Kant, rests on empirical intuitions, which must be the correlates of singular terms. For purely intellectual discourse, where the difference between intuition and concept is not a difference in kind, Kant might well take a more Leibizian line.

The second relatively clear point is that the Fregean refinement of our central notion — the notion of a concept — is indeed derived from a picture that goes back to Kant and Leibniz. Frege, famously, defines a concept as a special sort of function,[39] continuing, I believe, the dispositional Leibnizian account and the Kantian view that a concept is a "rule." And though their language is still mentalistic, the Leibniz-Kant drive to divorce the semantics of a concept from its particular origin in some person's conscious life is a forerunner of Frege's more radical anti-psychologism.[40]

But on some other predication related issues there is a less straightforward connection between modern philosophers of language and the views of Leibniz and Kant. I shall — again briefly — sketch two: the question of whether Frege is an "idealist" in the mold of Kant and Leibniz, and the issue of Kantian and modern criteria of singularity.

i. *Frege's idealism*

The question here is whether Frege is a straightforward classical realist about truth and about certain classes of objects, or whether he is (as Hans Sluga suggests) a linguistic and logical idealist in the tradition of Leibniz and Kant.[41] I divide this question into three separate

issues, which I call "referential truth," "humanism," and "definitional existence."

We have seen that both Kant and Leibniz use a relation among representations (be they intuitions or concepts) to define the notion of "agreement" which lies at the heart of elementary truth. Frege, of course, rejects this mediation of truth by mental states. For him, singular terms refer to objects and general terms to concepts. The truth of a simple predicative judgment depends only on the relation between the referents of its terms. So if we take this referential theory of truth as the mark of semantic realism [42] then we shall have to say that Kant and Leibniz are indeed not realists, and that Frege most certainly is.

I myself wonder, though, how true a mark this is of semantic realism and anti-realism. It seems to me that a better account of semantic realism revolves on what I above called "humanism": introducing characteristically human factors into the definition of elementary truth. Clearly, when our concern is empirical discourse, then these "human" factors must be epistemic. So a semantic humanist will take perception, perceptual evidence, and other epistemic conditions to be factors not merely in the knowledge of any given empirical judgment but in the actual truth of that judgment.[43] A realist will resist that humanization. Leibniz is without question a realist on this issue — as we have seen. While Dummett — and Kant too, I have argued — are humanists.

What of Frege? Well, in his own words: "Being true is different from being taken to be true."[44] And Russell as well says that truth is "a wider conception than knowledge." [45] There is no humanism here, in either of them.

As I said, I think that humanism is a better mark of semantic anti-realism than is the question of reference. For humanism offers a logical litmus test to distinguish realists from anti-realists, while reference does not. It is precisely their "humanism," the fact that they tie truth to humanly accessible knowledge, that leads Dummett and Kant to adopt a demonstrably non-classical logic. And it is precisely their anti-humanism that leads Leibniz and Frege to what we call classical logic.

The third idealistic issue, the one concerning existence and definition, is more metaphysical. When you define a concept (be it general or singular) you set out the *Merkmale* into which the concept can be analyzed. In the singular case, you might set out only sufficiently many *Merkmale* to distinguish the associated object from other objects, without fully exhausting the individual concept. That is what Leibniz calls a *nominal* definition. A *real* definition, by contrast, will set out all the *Merkmale*, or at least sufficiently many to determine the possible and even actual existence of the object. The possibility of real definition establishes what R.M. Adams calls the "causal structure" of individual concepts.[46]

This seems to me to provide a good mark of metaphysical realism. A metaphysical realist will deny the possibility of real definition in this sense. Conceptual manipulations, the realist will insist, cannot establish that an object exists. By this criterion Leibniz is an antirealist, and Frege once again comes out to be a realist.[47]

Kant interestingly falls between the two poles on the question of the criteria of existence. He insists adamantly that concepts alone cannot establish existence. That is the famous heart of his refutation of the ontological argument.[48] And indeed, it is also his ground for claiming that empirical individual concepts are not subject to [real] definition.[49] But following his semantic humanism, he also asserts that existential judgments are true only by virtue of their connection to perception. And from this he concludes that it is "sensation" that provides the mark of "reality" for any empirical object. So while empirical existence cannot be determined by concepts — and certainly not by their definitions, — it is still not independent of our human mental states.

ii. *Recognizing singularity*

In section (II.iv) I suggested that the thesis of the first antinomy should be symbolized as $\exists x \forall y_{\neq x} Fxy$, and the antithesis as $\forall x \exists y F_{yx}$. I then pointed out that Kant's arguments establish (for both the realist and idealist) the falsity of each and that these conclusions are consistent with the intuitionistic but not the classical truth of $\forall x \neg \neg \exists y Fxy$.

There is, however, another possible reading of this piece of the *Critique of Pure Reason*. On the alternative reading we can take finitude as a simple predicate (symbolized by, say, "G"), and we can view the term *the world* as a singular term (abbreviated "w) which refers to the set of all spatially occupied locations. We would then say that the Antinomy arguments show that this singular term, "the world", lacks reference, and $(Gw \vee \neg Gw)$ fails simply because of this failure of reference.

This is the interpretation suggested by Gordon Brit-

tan.[50] And it does apparantly fit Kant's own summary at A506/B534:

> If the world is a whole existing in itself, it is either finite or infinite. But both alternatives are false (as shown in the proofs of the antithesis and thesis respectively). It is therefore also false that the world (the sum of all appearances) is a whole existing in itself.

If we were to accept this reading, then that would put Kant on the Fregean side of a famous issue raised by Russell about the criteria for singularity. Russell claims that the only legitimate singular terms (the "logically proper names") are expressions which refer to objects with which we are directly acquainted. All other alleged singular terms must be analyzed as combinations of general terms and quantifiers. Russell held this because he believed that only referring terms are truly singular and because he had a strict knowledge-by-acquaintance view of reference.[51] Frege, by contrast, did not allow the issue of actual reference to determine whether a linguistic expression is to count as singular. In particular he certainly allowed the grammatical possibility (in natural language at least) that there should be non-referring singular terms. He thinks that literature is full of such empty names and descriptions. Indeed, he argues that in these cases the sentences in which these terms occur will lack truth value altogether.[52] And that indeed fits the second reading of Kant's argument in the "Antinomy."

However, I think this is not the best reading. Indeed, I think that Kant's views are a modification of Russell's side; at least as they regard empirical discourse. For one thing, you can see that Kant speaks of the thesis and antithesis as false outright and not as truthvalueless. But, more tellingly, if we understand the antinomy in the suggested way then Kant's summary would contain a massive *non sequitur*. For he goes on, in the continuation of the passage I just quoted, to conclude from the fact that the world is not a thing in itself that

> it then follows that appearances in general are nothing outside our representations — which is just what is meant by their transcendental ideality.

But of course, the claim that an abstract singular term like "the world" lacks reference does not entail the claim that ordinary empirical objects (appearances," in Kant's terms) must be "nothing outside our representations."

That is why I prefer to analyze Kant's claims about "the world" in a quantified idiom. Kant's criterion of linguistic singularity is simply the possibility that there might be a corresponding intuition. In the empirical case, that criterion amounts to the possibility of perception. And that is close to the Russellian position.

It is, however, not identical to Russell's view. For requiring the possibility of an intuition is not the same as requiring an actual intuition. Connection to an actual intuition is required in order to assert that the designated object exists, but it is not required for establishing the singularity of the designating expression. Thus Kant, unlike Russell, does allow the possibility of non-referring empirical singular terms.

In purely intellectual discourse — in the case, that is, where there is no distinction in kind between expressions related to concepts and those bearing intuitions — in this case the Fregean criterion for singularity can well suffice.

So if we say that Kant attempted in the "Antinomy" to show that the logic and grammar of intellectual discourse cannot apply to empirical discourse, then we can understand both parts of his summary at A506/B534. For since he has established that an intuition of "the world" is impossible in principle, he has established both that "the world" (as a putative object) does not exist and that judgments about the world must be understood as at best general judgments (i.e., judgments which we would symbolize using quantifiers). This is how we should understand his claim that the world is not a "whole existing in itself." But part and parcel of this new logic and grammar for empirical discourse is the claim that human perception is the criterion of existence. And that is how we should understand his conclusion that empirical objects "are nothing outside our representations."

It is inevitable that we read the history of philosophy through the lens of our own current philosophical interests; and it is always tempting to apply the latest methods to past masters. One of the things I have tried to show in this sketch is how beneficial such readings and applications can be. Our modern semantic outlook has helped us understand the relation between important themes in Leibniz and in Kant. And twentieth century machinery from logic and the theory of elementary predication has helped us interpret some subtle Kantian arguments.

But I also hope to have shown some constraints on this interpretative procedure. Kant and Leibniz do not fall into any current camp on the definition of elementary predicative truth, and consequently they do not

parse the issue of realism the way we do today. And though Kant might well advocate a logic akin to intuitionism for empirical judgments, his reasons are importantly different from Dummett's. Finally, his criteria for singularity do not fit neatly into either of the classic positions on this issue.[53]

In the end, all of these differences arise because Leibniz and Kant worked within a "paradigm" in the theory of truth and predication that is different from our own. I have begun to show, I hope, that the two paradigms are not at all incommensurable. But we must always remember that there is no simple, direct translation between them.

Notes

[1] *New Essays Concerning Human Understanding* (A. G. Langley, tr., Macmillan, 1896), Ch. V, sec. 2.

[2] "Idea" is Leibniz's generic term for a mental representation whether it is consciously present or not. A "concept" is an idea which has been consciously entertained. However he himself often ignores this distinction.

[3] "As a matter of fact our soul does have within it the disposition to represent itself any nature or form when an occasion arises for thinking of it. I believe that this disposition of our soul, insofar as it expresses some nature, form or essence, is properly speaking the idea of the thing, which is in us and is always in us whether we think of it or not" (*Discourse on Metaphysics*, XXVI).

[4] Raili Kauppi, in *Uber die Leibnizsche Logik*, (*Acta Philosophica Fennica*, Fasc. XII, 1960, pp. 248 ff.) suggests that this process of resolution generates a partial ordering on the field of all concepts.

[5] "It is the nature of an individual substance or complete being to have a concept so complete that it is sufficient to make us understand and deduce from it all the predicates of the subject to which the concept is attributed" (*Discourse on Metaphysics*, VIII).

[6] "The predicate or consequent therefore always inheres in the subject or antecedent. And, as Aristotle, too, observed, the nature of truth in general or the connection between the terms of a proposition consist in this fact" ("First Truths," in G. W. Leibniz, *Philosophical Papers and Letters*, L. E. Loemker (ed.), Reidel, Second Edition, 1956, pp. 267—270).

[7] "Surely since God can form and does in fact form this complete concept which includes enough to explain all the phenomena that come to me, such a concept is possible and is the true complete concept of what I call myself, in virtue of which all my predicates pertain to me as their subject. This could be proved just the same, then, without mentioning God, except insofar as it is necessary to indicate my dependence . . ." (Letter to Arnauld, July, 14, 1686, Loemker, p. 335).

[8] "When my mind understands at once and distinctly all the primitive ingredients of a conception then we have intuitive knowledge. This is extremely rare as most human knowledge is only confused or indeed assumed (*Discourse on Metaphysics*, XXIV).

[9] In his paper "Predication, Truth and Transworld Identity" (in J. McGuire and J. Bogen (eds.), *The Way Things Are*, Kluwer Acad. Publ., 1980, p. 250) R. M. Adams claims that Leibniz's containment theory is "anti-semantical." He claims this because for Leibniz containments and negations are mechanically determined and because truth and falsity do not depend on objects. Adams' conclusion is wrong. Semantics can be mechanical. (Monadic quantificational semantics is mechanical.) And determining the ordering and interreducibility of concepts is actually the heart of meaning theoretic semantics. But the bite of Adams' second point is nonetheless well taken: For Leibniz, reference to objects is not the basic component in determining the truth of a judgment. The properties of an object are a consequence of the truth of judgments about the object and not *vice versa*.

[10] Leibniz is not alone, of course, in taking logic to be an ontological affair. Husserl, for instance, speaks of logic as "formal ontology."

[11] I do not want to enter into the debate (initiated by Russell and Couturat) about whether Leibniz derived his metaphysics from his logic or *vice versa*. I am here merely noting the fact that these principles can be derived from the claim that individual objects are complete.

[12] "For if anyone were capable of carrying out a complete demonstration by virtue of which he could prove this connection of the subject, which is Caesar, with the predicate, which is his successful enterprise, he would bring us to see there a reason why he resolved to cross the Rubicon rather than to stop, and why he gained instead of losing the day at Pharsalus, and that it was reasonable and by consequence assured that this would occur . . ." (*Discourse on Metaphysics*, XIII).

[13] Thus, to use a famous example, two apparantly distinct drops of water which have no conceptual difference cannot in fact be distinct. In particular, because of Leibniz's relational theory of space, he will not allow us to use their alleged spatial separation to distinguish the droplets. The space between them is a corollary of their difference, and not *vice versa*.

[14] It is of course mediated by the work of people like Crusius and Wolff. But, given the tradition in which these intermediaries are working, that fact highlights rather than diminishes Kant's Leibnizian roots.

[15] "In all subsumptions of an object under a concept the representation of the object must be *homogeneous* with the concept; in other words, the concept must contain something which is represented in the object that is to be subsumed under it. This, in fact, is what is meant by the expression, 'an object is contained under a concept' " (A137/B176).

[16] "But every *thing*, as regards its possibility, is likewise subject to the principle of *complete* determination, according to which if *all the possible* predicates of *things* be taken together with their contradictory opposites, then one of each pair of contradictory opposites must belong to it" (A571—2/B599—600). See also sections II(v) and II(vi) below.

[17] "Humanity [as an idea] in its complete perfection contains not only all the essential qualities which belong to human nature and constitute our concept of it — and these so extended as to be in that complete conformity with their ends which would be our idea of perfect humanity — but also everything which, in addition to this concept, is required for the complete determination of the idea. For of all contradictory predicates one only (of each pair) can apply to the idea of the perfect man" (A568/B596).

[18] "If, in following up this idea of ours, we proceed to hypostatise it, we shall be able to determine the primordial being through the mere concept of the highest reality, as a being that is one, simple, all-sufficient, eternal, etc. In short, we shall be able to determine it, in its unconditioned completeness, through all predicaments. The Concept of such a being is the concept of *God*, taken in the transcendental sense . . ." (A580/B608).

[19] This can interpret Kant's claims about objects conforming to our knowledge in the famous "Copernican Revolution" passage at Bxvi—xvii.

[20] I am using the word "perception" in the standard way rather than the more general sense of "representation with consciousness (*perceptio*)" of A320/B376.

[21] "This (*cognitio*) is either *intuition or concept*. The former relates immediately to the object and is single, the latter refers to it mediately by means of features which several things may have in common" (A320/B377).

[22] Notice that Kant has a looser conception of unity than Leibniz. For Leibniz the unity of an individual object comes from the completeness of its concept. This, recall, is the only legitimate source of unity that is sufficient to constitute an object. It is what makes it, in Fregean terms, "saturated." This completeness is also exactly the property that empirical concepts and intuitions lack. But Kant simply says that, in spite of their incompleteness, singular empirical concepts are perfectly good concepts, and empirical perception is sufficient to adequately represent an object and to provide unity.

[23] See my "Autonomy, Omniscience and the Ethical Imagination," in Y. Yovel (ed.) *Kant's Practical Philosophy Reconsidered*, Kluwer Academic Publ., 1989.

[24] "(For empirical objects) even if there is no difference whatever as regards the concepts, difference of spatial positions at one and the same time is still an adequate ground for the *numerical difference* of the object, that is of the object of the senses. Thus in the case of two drops of water . . . the mere fact that they have been intuited simultaneously in different spatial positions is sufficient justification for holding them to be numerically different" (A263—4/B319—20). See note 13 above.

[25] For a précis of intuitionistic logic and semantics see D. van Dalen, "Intuitionistic Logic," in D. Gabbay and F. Guenther (eds.), *Handbook of Philosophical Logic*, v. III, Kluwer Academic Publishers, pp. 225—339.

[26] E.g., $\forall x \forall y[Fxy \rightarrow \neg Fyx]$ and $\forall x \forall y[Fxy \rightarrow \neg(x=y)]$.

[27] "The complete determination is thus a concept, which, in its totality, can never be exhibited *in concreto*. It is based upon an idea, which has its seat solely in the faculty of reason — the faculty which prescribes to the understanding the rule of its complete employment" (A573/B601).

[28] "[O]n the one hand, reality is independent, not necessarily of thought in general, but only of what you or I or any finite number of men may think about it; and . . . , on the other hand, though the object of the final opinion depends on what that opinion is, yet what that opinion is does not depend on what you or I or any man thinks. Our perversity and that of others may indefinitely postpone the settlement of opinion; it might even conceivably cause an arbitrary proposition to be universally accepted as long as the human race should last. Yet even that would not change the nature of the belief, which alone could be the result of investigation carried sufficiently far; and if, after the extinction of our race, another should arise with faculties and disposition for investigation, that true opinion must be the one which they would ultimately come to. 'Truth crushed to earth shall rise again,' and the opinion which would finally result from investigation does not depend on how anybody may actually think. But the reality of that which is real does depend on the real fact that investigation is destined to lead, at last, if continued long enough, to a belief in it." (C. S. Peirce, "How to Make Our Ideas Clear", in *Collected Papers*, Vol. V, 248—271).

[29] "To call an appearance a real thing prior to our perceiving it, either means that in the advance of experience we must meet with such a perception, or it means nothing at all" (A493/B522).

[30] See note 16 above.

[31] Indeed he explicitly derives this from the "*universality* of the principle of excluded middle" (A572/B600 n.).

[32] The principle I am advocating is $\exists x(x=t) \rightarrow (Pt \vee \neg Pt)$, which deviates from standard intuitionistic logic.

[33] ". . . this idea of the *sum-total of all possibility*, in so far as it serves as the condition of the complete determination of each and every thing, is itself undetermined in respect of the predicates which may constitute it . . ." (A573/B601).

[34] See Dummett's essay "Realism" in *Truth and Other Enigmas*, Harvard University Press, 1978, pp. 145—165.

[35] In other publications I have used Kant's affinity with assertabilism to defuse the accusation of phenomenalism. Here, however, I want to explain rather than reject his phenomenalist language.

[36] See also A191/B236.

[37] "Predication joins a general term and a singular term to form a sentence that is true or false according as the general term is true or false of the object, if any, to which the singular term refers." W. V. Quine, *Word and Object*, (MIT Press, 1960), p. 96.

[38] "You say there are two kinds of concepts, some representing a single unified something from which nothing can be separated without destroying the whole; . . . The other kind of concept may represent two or more things. This is a little obscure to me. Surely every concept or definition is such that you cannot remove anything without destroying the whole definition; . . ." Leibniz, Letter to DeVolder of June, 23, 1699.

[39] See "Function and Concept" in P. Geach and M. Black (eds.) *Philosophical Writings of Gottlob Frege*, Blackwell's, 1966, pp. 21—41.

[40] See for instance Frege's review (in *Zeitschrift für Philosophie und Kritik*. v. 103, 1894) of Husserl's *Philosophie der Arithmetik*.

[41] See H. Sluga, *Gottlob Frege* (Routledge and Kegan Paul, 1980), pp. 58 ff.

[42] As does (e.g.,) Hilary Putnam in *Meaning and the Moral Sciences* (Routledge and Kegan Paul, 1978).

[43] Nelson Goodman (in e.g., *Ways of Worldmaking*, Hackett, 1978) and Hilary Putnam (in *Reason, Truth and History*, Cambridge University Press, 1981) have suggested that we must also add considerations concerning human values and interests to the humanistic semantics.

[44] *Grundgesetze der Arithmetik*, v. I, p. xv.

[45] *An Inquiry into Meaning and Truth* (George Allen and Unwin, 1940), p. 246.

[46] "The containment of Adam's sin in Adam's individual concept is not an *alternative* to the containment of the sin in Adam himself. For Adam's individual concept represents Adam's permanent internal causal structure, so that whatever sin follows from the one follows from the other." R. M. Adams, *loc. cit.*

[47] "Here it is of importance to make clear what definition is and what we can reach by means of it. It is, it seems, often credited with a creative power; but really all there is to definition is that something is brought out, precisely limited and given a name. The geographer does not create a sea when he draws border lines and says: The part of the surface of the ocean delimited by these lines, I am going to call the Yellow Sea; and no more can the mathematician really create anything by his act of definition. Nor can we by a mere definition magically give to a thing a property which it has not got, apart from the property of now being called by whatever name one has given it. But that an oval drawn on paper with pen and ink should acquire by definition the property that, when it is added to one, one is the result, I can only regard as a scientific superstition. One might just as well make a lazy pupil dilligent by a mere definition" (*Grundgesetze*, v. I, xiii—xiv; translated in Geach and Black, *op. cit.* pp. 144—145).

[48] A592—602/B620—630.

[49] A728—729/B756—757.

[50] *Kant's Theory of Science* (Princeton University Press, 1978), pp. 39ff.

[51] See, for instance, Russell's "On Denoting" (*Mind*, 1905), "Knowledge by Acquaintance and Knowledge by Description," (*Proceedings of the Aristotelian Society*, 1910—1911), and his 1918 "Lectures on Logical Atomism" (reprinted in *Logic and Knowledge*, 1956).

[52] See "On Sense and Reference," in Geach and Black, *op. cit.* pp. 56—78.

[53] Nor, for that matter, do Kant's criteria coincide with any of the subsequent refinements of Russell's and Frege's views about the connection between singular predication and existence.

Duke University